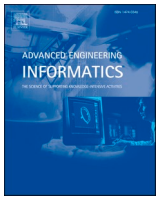
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Full length article



Digital twin-enabled collision early warning system for marine piling: Application to a wharf project in China

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ABSTRACT

The safety and effectiveness of marine piling construction are hindered by the display of imprecise spatial information and the potential for collisions due to the complexity of pile positions and attitudes. Digital twin (DT) can solve the above problems in piling construction because of their inherent three-dimensionality and real-time feedback. The present study proposes a system that utilizes the DT framework to construct a collision early warning system intended for marine piling. A five-dimensional model is used to introduce the overall architecture. The main twin objects are piles. To maximize the effectiveness of DT, the five-dimensional model is composed of four independently maintainable development modules. Consequently, a prototype of the system is created. The core of the system comprises an algorithm for pile positioning and an algorithm for collision early warning, especially for complex pile groups. Two distinct implementations are identified: one involves controlling the virtual with the real, while the other involves controlling the real with the virtual. Field experiments are conducted with the aim of establishing a benchmark for collision early warning during the construction process. The prototype of the system is capable of producing a simulated environment through a dual-layer mapping approach, which involves pile-boat mapping and early-warning feedback mapping. The presented system in a practical wharf project showcases both the capacity for three-dimensional visualization of collision warning and the ability to provide feedback for boat control. The research enables more intuitive and effective pile positioning.

1. Introduction

Pile foundations have emerged as a crucial component in the development of viaducts, high-rise buildings, offshore ports, and water transport wharves. Based on the characteristics of the construction environment, piling can be divided into two categories: land piling and marine piling. The conventional approach for piling involves the utilization of a total station, or theodolite, which is measured by construction personnel [1]. In contrast to piling on land, marine piling is vulnerable to external influences, including the swinging of the piling boat, fluctuations in the water surface, and the effects of wind [2]. The efficiency of manual construction positioning is low, and its precision is limited [3].

Global navigation satellite systems (GNSS) and real-time kinematic (RTK) positioning are getting more mature, and their costs are steadily decreasing. These tools find extensive application in the domain of civil engineering, including but not limited to construction machinery control [4,5], high-rise buildings [6], bridges [7], and dam structural health monitoring [8]. The level of accuracy in positioning has reached a resolution of millimeters. The initial implementation of the GNSS in piling was documented by Seward et al. in their publication [1]. The automatic land drilling positioning was achieved through the utilization of the advanced PC1O4 processor. The positioning of piles on land was investigated by Takeshi et al. [2,9,10] through the utilization of a line laser range finder. Yao et al. [3] and Tserng et al. [11] have presented models for marine piling that address the challenge of computing the coordinates of the pile center in real time. The proposed model utilizes a combination of multiple GNSS receivers, inclinometers, and range finders to develop diverse algorithms for pile positioning calculations. The algorithms use three types of coordinate systems, namely geodetic coordinate system, engineering coordinate system, and ship-fixed coordinate system. The geodetic coordinate system is used to measure the

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position of a certain point on the earth, that is, the longitude, latitude, and elevation coordinates received by the GNSS receivers [12]. The engineering coordinate system is used to measure the position of a certain point in the construction area. Different engineering projects have different engineering coordinate systems [13]. The ship-fixed coordinate system is used to measure the position of a certain point relative to the hull. Different ships also have different ship-fixed coordinate systems. The projection transformation method can be used to convert the geodetic coordinate system into an engineering coordinate system, allowing GNSS to be used in engineering. Using the three-dimensional datum transformation model [14], the ship-fixed coordinate system can be converted into an engineering coordinate system, so that the boat can be used in marine piling construction. The algorithms are utilized to determine the ship-fixed coordinates that describe the pile center. These coordinates are subsequently converted into engineering coordinates through the utilization of a parameter that facilitates the transformation of ship-fixed coordinates into their corresponding engineering coordinates. The utilization of charge coupled device (CCD) cameras in the above algorithms was recently investigated by Yao et al. [15]. A two-dimensional software for marine pile positioning was developed utilizing the pile positioning model mentioned above. This software has been adopted by Shanghai Huace Navigation Technology Ltd., Guangzhou Zhonghaida Satellite Navigation Technology Co., Ltd., and China Communications Construction Co., Ltd. for the purpose of pile positioning in China's offshore foundation construction. It directs the piling boat to detect the pile location and torsion angle, and it controls the pile frame to tilt forward and backward to establish the pile slope ratio. Thus, it achieves the positioning of marine piling without the involvement of a total station or theodolite.

The algorithm for pile positioning utilizes a combination of GNSS receivers, inclinometers, and rangefinders to facilitate accurate positioning across a variety of working conditions. Although positioning technology has been improved, the two-dimensional thinking that originated from manual drawings still continues. The pile is represented as an ellipse at the point where the engineering design elevation plane intersects with the axis of the pile. This replacement of the pile results in the loss of information regarding the three-dimensional staggered arrangement of the piles in space. Practical engineering experience in marine piling construction tells us that the layout of the pile body is a staggered arrangement of straight piles and inclined piles. Considering that the objects below the water surface cannot be foreseen during construction, the positioning challenge is further increased. Also, it is challenging for construction personnel to acquire precise spatial position data for complex construction conditions. They must rely on their experience to steer the boat. In addition, unexpected pile positions and attitudes may occur during construction. There are two reasons for this unexpectedness. On the one hand, while the design phase has successfully addressed the potential for pile collision, the limitations imposed by the vessel's configuration and the load-bearing capacity of the pile framework during construction have not been taken into account. On the other hand, the physical act of construction can lead to a minor deviation in both the torsion and inclination angles of the pile. The angle deviation is determined by the inclination of the pile, the frame of the boat, and the weight of the pile. Small changes in angle can cause large changes in the position and attitude of the pile in three-dimensional space. Therefore, the current two-dimensional marine piling positioning software is unable to intuitively and in real-time map the actual construction conditions for construction operation feedback. The actual construction pile position does not match the one specified in the design. The difficulty of construction and the risk of collision increased. This collision refers to the collision between two pile bodies. The collision between piles under construction and piles that have been completed is something that should be avoided during construction and is the main purpose for establishing the digital twin (DT).

Therefore, based on the current pile positioning technology, marine piling urgently needs intuitive and precise technology to reduce the challenges and risks of pile sinking construction. The DT has natural three-dimensional properties. It is a real-time twin of reality. Grieves first proposed the concept of the DT as a "product life cycle management model" [16]. Despite the fact that many research results have surfaced after two decades of development and implementation, 96% of them have emerged since 2016 [17], with 50% of these results pertaining to manufacturing and production [18]. Only 5% are associated with the built environment [19]. The absence of comprehensive theoretical frameworks and construction techniques for unified DT models is evident, even within the context of manufacturing and production. Tao et al. described in detail the essential technologies, enabling technologies [20,21], and instruments for establishing DT models [22]. Jia et al. [23,24] constructed a DT model with multiple scales and scenes in two sections. In recent years, hardware and algorithm iterations have resulted in the emergence of new technologies like 5G [26], the Internet of things [27], and mixed reality [28], which have been integrated with DT to adapt to modern Industry 4.0 development with the concept of "combining the virtual and the real, controlling the real with the virtual." The implementation of this technology is progressively being observed in various sectors such as manufacturing and production [29,30], construction, energy, and medical domains [31]. The civil engineering is currently in the early stages of conducting research on DTs, which means implementing a DT is extremely challenging [25]. In addition to the complexity of DTs mentioned above, there exists some confusion and its potential overlap with similar technological research, including building information modeling and cyber-physical systems [25].

DTs can provide real-time mapping of the real physical world. This is important for marine piling construction. Because it requires intuitive and precise technology to solve three problems: (1) two-dimensional representation that lacks spatial information, for example, ellipse replacement for inclined piles; (2) the position of piles under the water surface is not visible; (3) actual construction machinery limitations and the physical act of construction cause it to deviate from design and increase the risk of collisions. The utilization of intuitive three-dimensional displays and real-world control over virtual counterparts has proven to be an effective solution for piling. At this stage, marine piling construction applies GNSS receivers, inclinometers, and rangefinders. An increasing volume of data has been produced, encompassing both structured data, such as time series that incorporate positional data, and unstructured data, such as audio and video. These data are important and necessary conditions for building a DT system. Real-time and rich data can establish a practical DT-enabled collision early warning system for marine piling. The construction of a DT and its paradigm consists of at least three essential components: the entity, the twin virtual body, and the twin data [21]. This study represents the inaugural application of the DT in the context of collision early warning for marine piling.

The aforementioned issues have prompted the investigation and development of algorithms for pile positioning and collision early warning in challenging pile construction. One algorithm offers precise positioning, while the other algorithm supplies collision warning data during the initial piling phase. Subsequently, a virtual model that accurately corresponds with the entity in real-time is generated to simulate the process of piling. Piles serve as main DT objects. At the actual construction site, the DT-enabled collision early warning system was used. The utilization of the system has the potential to facilitate the management of the boat and pile frame through a simulated model that provides timely feedback. The construction process is systematically monitored and achieved a collision early warning between piles under construction and completed piles. At the same time, the intuitiveness and accuracy of pile positioning are enhanced.

2. Framework of the system

2.1. Basic process and collision probability of piling

The marine pile positioning is achieved by first striking the pile body with the boat and then driving it with a pile hammer. The primary hull, cockpit, and pile frame make up the main part of the boat. The cockpit regulates the movement of the hull, pile hanging, and pile hammering. The holder and the pile hammer make up the pile frame. The hammering and hanging are handled by the pile frame. There are four significant steps in the piling process:

(1) Hanging. The pile frame leans slightly forward to lift the pile body. As the holder is stabilized, gradually move the hull closer to its intended position. The pile is then locked by the holder to reduce shaking.
(2) Adjusting angles. The angles are determined by the designated torsion, slope ratio, the actual weight, and the pile frame bearing capacity. During this time, the pile positioning algorithm adjusts the pile's position in accordance with the design.
(3) Fine positioning. The pile is gradually fine-tuned to the design position. At the same time, move the boat so that it is at a specified safe angle from the cable. Then, sink the pile into the water until it reaches the bottom and stabilizes the hull for a period of time.
(4) Start piling. The diesel hammer on the pile frame is placed at the proper height so that the force of the pile is even. Evacuate workers after installation and plumbing. The cockpit gives instructions to complete the construction via pile driving.

Fig. 1 shows the four important steps of the piling process, where changing the angles and fine-tuning the position in the middle photograph can easily lead to a collision. It is also the fundamental reason for establishing a DT-based collision early warning system for marine piling. The construction difficulty increases when piles are relatively dense and arranged in a staggered manner. Suppose the bearing capacity of the pile frame is insufficient to elevate the pile to the design slope ratio. In this case, there will be a non-negligible deviation between the constructed and designed pile positions.

2.2. Challenges of collision

The main function of the DT-enabled collision early warning system is to indicate possible collisions and then avoid them. The collision here refers to the collision between piles under construction and completed piles. In marine piling construction, especially in wharves, the arrangement of piles is relatively complicated. Straight piles and inclined piles are arranged crosswise to meet the bearing capacity requirements. In the design stage, collision has been avoided, but the hull, construction personnel, and construction environment have not been considered. They are factors that determine whether the final project conforms to the design. Based on actual construction experience, our designs may be subject to change due to insufficient geological survey. For example, increase pile density or increase the inclination angle of inclined piles to improve the overall bearing capacity of a wharf. Embedding the piles to be constructed around the piles that have already been constructed increases the challenges of construction and the possibility of collisions. It is rare when designs change. What is more common is that construction personnel relax the implementation of design requirements due to the constraints of the hull shape and construction environment. For example, because the angle of the inclined pile is too large, the weight of the pile body exceeds the bearing capacity of the pile frame of the boat, making it impossible to lift the pile body during piling. In this situation, the angle of the inclined pile must be reduced. Therefore, construction deviations occur. The inclination and position that will not cause collision are no longer the design values, when the construction of the next pile is carried out. Therefore, both inclination and displacement cannot be warned before collision takes place. The closer the piles are, the more challenging it is to avoid collisions.

However, two-dimensional drawings of pile bodies and hulls are used to guide piling construction. Planar images inevitably lose spatial information, so the possibility of detecting collisions is very small. Almost all solutions rely on the experience of construction personnel. Any carelessness may cause a collision. Therefore, collisions are difficult to avoid relying on construction personnel. Avoiding collisions is challenging.

2.3. Closed-loop data flow in piling

The acquisition of monitoring data takes place initially during the construction phase. The global positioning data obtained through the

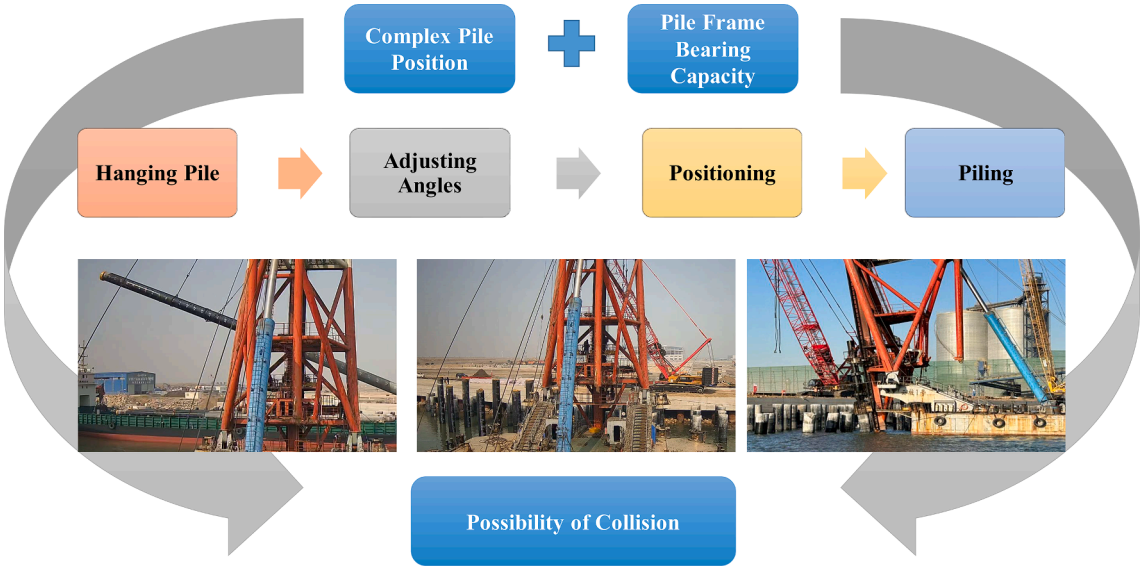


Fig. 1. The significant four steps and on-site pictures of pile positioning and the causes of collision risk.

GNSS receiver provides information on the longitude, latitude, and altitude of the receiver. Primary data is crucial for accurately determining the positioning of piles. The rangefinder is utilized to determine the distance between the measuring instrument and the pile. The inclinometer is utilized for the direct measurement of the forward and backward inclination of the pile frame, with the aim of achieving slope ratio specifications. The inclinometer is utilized to gauge the roll and trim of the hull, with the aim of ascertaining the extent of oscillation and rectifying minor discrepancies. The pile positioning algorithm and collision early warning algorithm rely on calculations based on global positioning data, distances, and angles. Consequently, in the digital-enabled collision early warning system, the data regarding the collision between the pile that is currently under construction and the one that has already been constructed is displayed to the construction personnel in both two and three dimensions. Simultaneously, the system interface receives early warning information and hull operation data to facilitate collision avoidance and steer the vessel's trajectory.

Fig. 2 shows the flow of data during pile positioning, encompassing the acquisition of global positioning data, distances, and angles, as well as the analyses and calculations of positioning and collision warning. Additionally, real-time construction information, two-dimensional and three-dimensional graphics rendering displays, early warning information, and boat operation information are included in this process. The green arrows in the figure illustrate this data flow that is a closed-loop data flow for pile positioning. The system achieves the integration of real-time mapping and data fusion between the physical and virtual environments.

2.4. Digital twin-enabled collision early warning system for marine piling

This system combines three fundamental elements, namely a DT entity, a twin virtual body, and twin data, with the requirement for collision early warning in pile positioning under complex pile positions. The main purpose of the DT is to prompt and avoid collisions between pile bodies, so the main object of the twin is piles. The proposed model shown in Fig. 3 is a five-dimensional model of marine piling, comprising a physical environment, a virtual environment, a feedback service, a connection, and twin data.

(1) Physical environment

The five-dimensional model is based on the physical environment and includes GNSS receivers, inclinometers, rangefinders, terrain, water surface, and physical entities of pile and boat. Our analysis involves the examination of torsion and slope ratio, as well as the amplitude of water surface fluctuation. The accuracy of pile positioning is directly influenced by factors such as the degree of terrain undulation, tide level, pile attitude, and boat position. Piles refer to both piles that are currently under construction and those that have already been constructed. In the design phase, particular emphasis is placed on spatial information such as torsion, slope ratio, and spatial location within the design elevation plane. For the boat, the focus is on swing and spatial position, especially the shape of the hull.

(2) Twin data

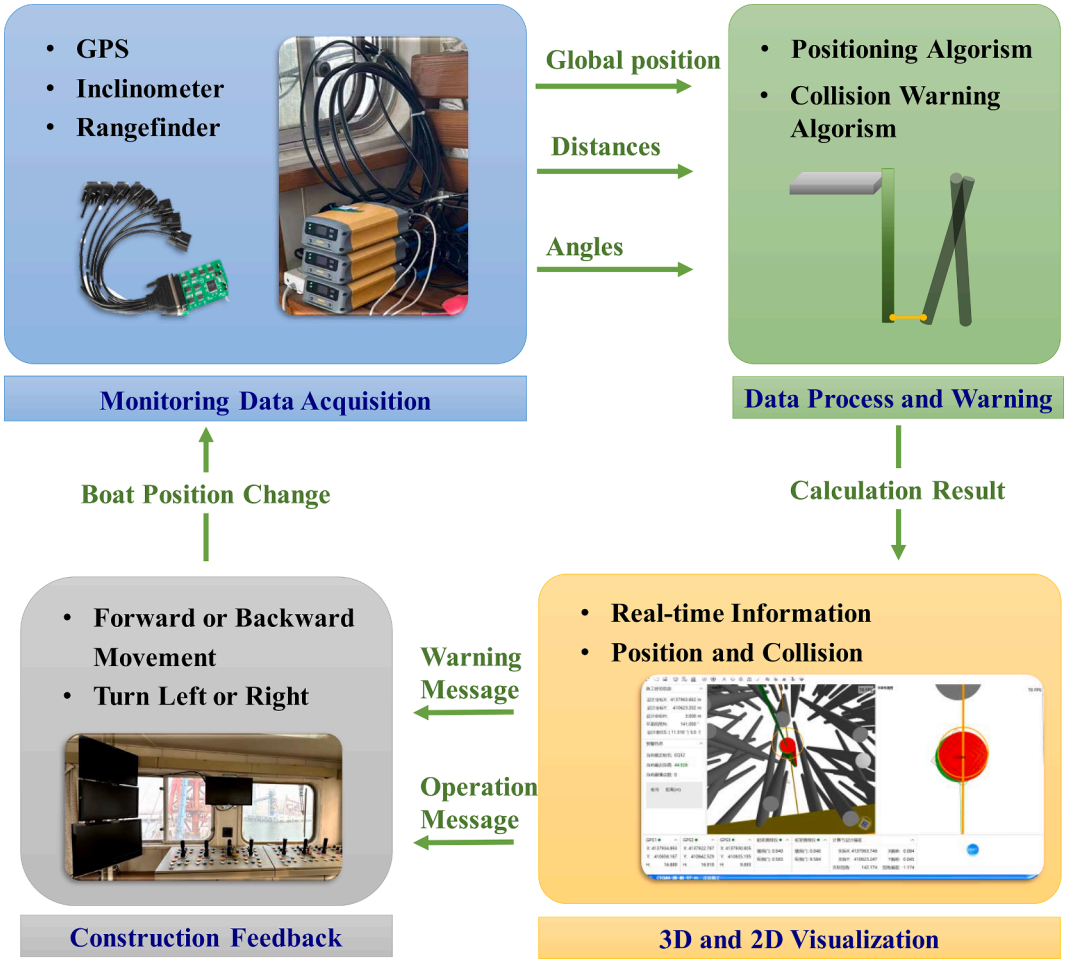


Fig. 2. Generation, collection, analysis, and processing of closed-loop data flow in pile positioning.

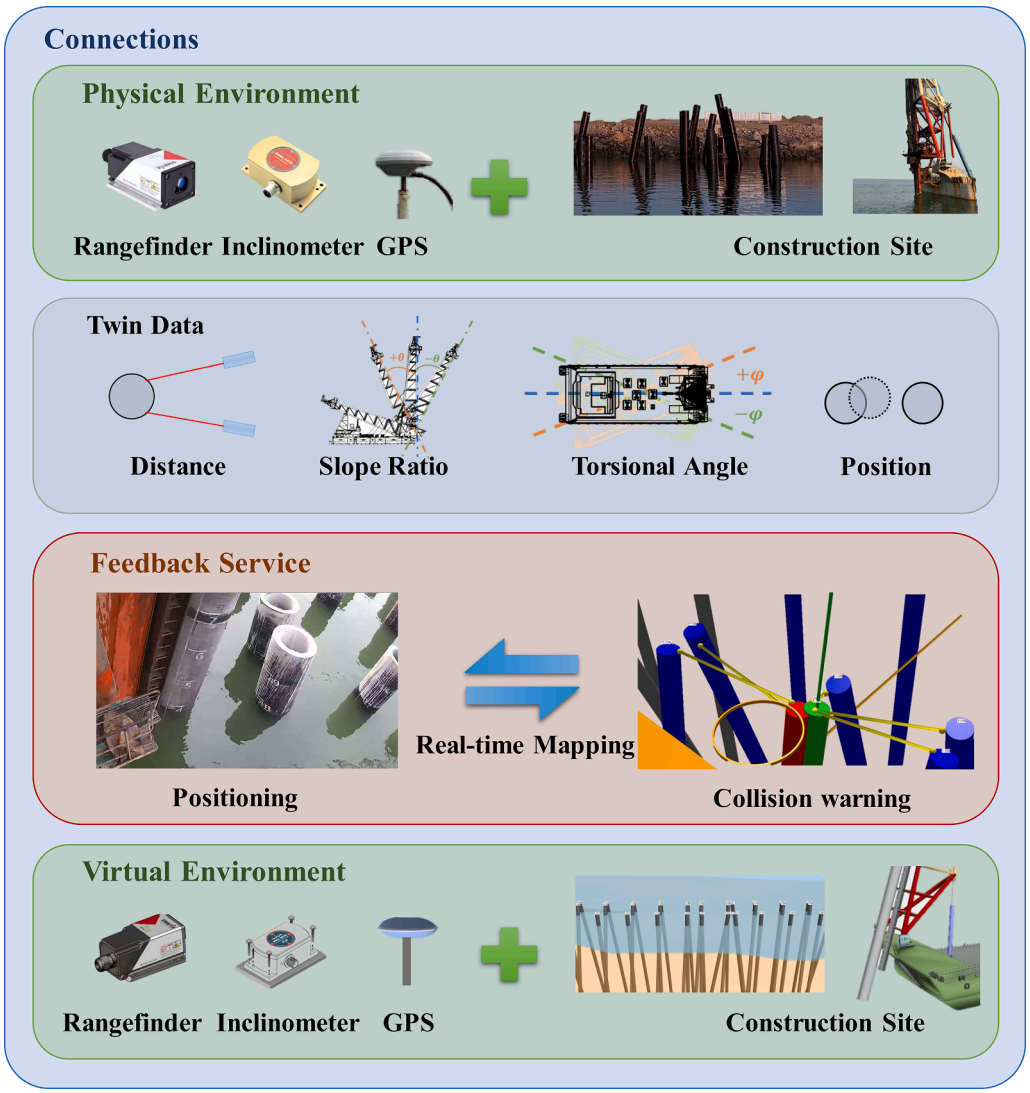


Fig. 3. Five-dimensional model of DT-enabled marine piling.

The basis of a five-dimensional model is twin data, which primarily consists of data regarding the physical environment, the virtual environment, and feedback service data. The physical environment data mainly includes the process of dynamic positioning, which is comprised of global positioning data, distance, and angles. Additionally, it describes the attribute information of several physical elements, including terrain, water surface, pile, and hull shape. The aforementioned data has the potential to accurately depict the current state of the physical environment. It is measured on-site and acquired via equipments. The virtual environment data mainly includes geometric models, physical models, and behavior models. The computing feedback service data encompasses various components such as algorithms, early warning feedback, database operations, and interactive data.

(3) Feedback service

The main body of the five-dimensional model is the feedback service. It supports three-dimensional modeling and human–computer interaction. Feedback services can be categorized into functional services and business services according to various requirements. Business services are built on functional services.

Functional services: The proposed tasks are as follows: (a) Perform the construction and rendering of models and virtual scenes for virtual environments; (b) Execute data cleaning, storage, packaging, mining, fusion, and analysis; (c) Provide interface services for connection as well as protocol services.

Business services: (a) Provide various monitoring methods for pile positioning, such as status display, information prompts, and operation feedback; (b) Provide real-time data for functional services; (c) Provide decision-making rules to construction personnel through human–computer interaction.

(4) Virtual environment

The virtual environment is characterized by its ability to replicate the geometric properties of the physical environment, including its shape, size, position, and other relevant parameters, through the use of three-dimensional modeling techniques. As such, it bears a strong resemblance to its physical counterpart. The utilization of model rendering technology enhances geometric modeling, while the incorporation of essential lighting, materials, and special effects imbues the virtual environment with a natural texture. Simultaneously, the ability to navigate through scenes is achieved through the development of a human–computer interaction interface and corresponding external input event response mechanisms.

(5) Connection

Real-time mapping of DT-enabled marine piling requires a reliable connection. The primary components of this system comprise the acquisition of real-time data through a diverse range of sensors, the mapping of real-time data onto the virtual model, the mapping of the actual physical environment onto the virtual environment, and the establishing of a connection between the physical and virtual environments. Twin data is formed through the utilization of conventional hardware interfaces and data transmission protocols, which are employed to facilitate the processes of data collection, transmission, and storage. Thus, a bidirectional linkage is established between the tangible object and the computational feedback mechanism. The utilization of GNSS receivers, inclinometers, and rangefinders facilitates the reflection of hull and pile data, thereby enabling the establishment of a correlation between the physical environment and twin data. The aforementioned fundamental connections enable the implementation of the pile positioning algorithm and collision early warning algorithm.

The DT-based collision early warning system for marine piling can be facilitated through the utilization of the above five-dimensional model. For piles, the main twin objects, the precise spatial attitude is indispensable data to the actual physical environment of the pile into a DT. These data were obtained with three kinds of instruments. GNSS receivers obtain the longitude and latitude coordinates during construction, the inclinometer obtains the inclination angle of the pile, and the rangefinder obtains the distance between the pile and the hull. Three kinds of instruments are used to convert piles into DTs. During the steps of adjusting the angle and positioning the pile, it is possible to generate a real-time map of the physical environment and receive collision warning feedback from the virtual environment. Upon receipt of the warning information, the construction personnel shall proceed to implement the system feedback operation in order to make the necessary adjustments to the boat position. At this point, the virtual environment will evaluate the collision risk in real-time and render the updated risk information, which can be used to visually locate the risk in order to prevent potential safety issues.

3. Procedures for pile positioning with collision early warning

A methodology has been devised to assess the suitability of the five-dimensional model, which aligns with the concept of controlling the virtual with the real. It comprises four distinct modules, namely the data management module, feedback module, motion drive module, and system management module. Each module may be developed and maintained independently.

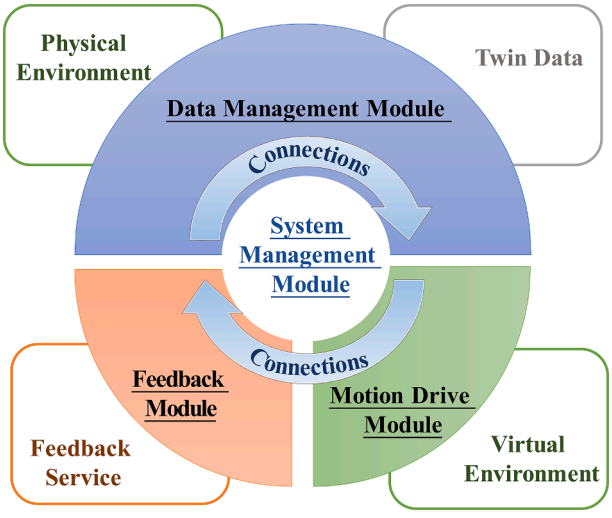


Fig. 4. Interconnection between the four modules and the five-dimensional model.

The interconnection among the modules of the aforementioned five-dimensional model is shown in Fig. 4. The physical environment of the method is accessed through the data management module, which primarily relates to different forms of instrument interfaces that represent physical reality, such as GNSS receivers, inclinometers, and rangefinders. Three kinds of instruments are used to convert piles into DTs. Additionally, it creates and stores twin data, such as dynamic process data and attribute data for physical elements. The motion-driven module is a virtual environment implementation that incorporates twin data real-time mapping, including real-time sensor data display, two-dimensional and three-dimensional graphic rendering, human–computer interaction, and other related functionalities. The feed back module that responsibles for command feedback is associated with the feedback service, which facilitates the implementation of the virtual control function. This function encompasses the pile positioning algorithm and collision early warning algorithm, both of which are designed to operate effectively in complex pile positions. The system management module coordinates the connection of all modules pertaining to the five-dimensional model.

3.1. Data management module

This module is responsible for governing the entirety of the system's data, including the refreshing of external data, driving model movement, and conducting data analysis. It comprises two distinct modules, namely a data transmission module and a data visualization module, as shown in Fig. 5.

The primary function of the data transmission module is to oversee the management of information obtained by the data portal while simultaneously addressing data requests from each module. The data is generated by real-time measurements of the instrument. GNSS receivers receive latitude and longitude coordinates, inclinometers measure the angle of the pile, and rangefinders measure the distance between the pile and the hull. These data are sent to the computer via RS232 or RS485 protocol. DT-enabled collision early warning systems can receive these data by setting the same protocol parameters. The software interface structure is defined by the Serial Port class in .NET. The Serial Port class contains many core methods for controlling hardware devices. It is the main way to realize twin data and data flow from physical to virtual. The information is transmitted to the program in the form of dictionary data pairs. After the above data is received, it is processed by other modules to form twin data. The Viewport3D class in .NET is utilized by the data visualization module to offer a display scene that effectively presents comprehensive details regarding the functioning of the construction site through an intuitive visualization. It depicts a hierarchical structure of terrain, water, piles, and vessel objects, which are organized and labeled for content display management.

3.2. Motion drive module

The motion driving module is responsible for controlling the virtual environment, including both dynamic and static data driving. Dynamic data drive pertains to the fluctuation of torsion angle, slope ratio, water surface, and spatial boat position. This ensures conformity with reality by promptly updating the data based on the twin data, thereby enabling control of the virtual and the real. The static data includes terrain, pile size, hull shape, and other three-dimensional model information related to a given scene.

The motion driven module is accountable for the construction of scenes in the program. The flow of this module is illustrated in Fig. 6 and primarily encompasses geometric modeling such as piles, boats, water, terrain, etc. In order to effectively execute collision warnings, it is imperative to utilize high-fidelity models. The high-fidelity model represents an enhanced version of the geometric model, while the realistic texture incorporates the lighting, materials, and special effects, as per the twin data-driven approach. The field of human–computer

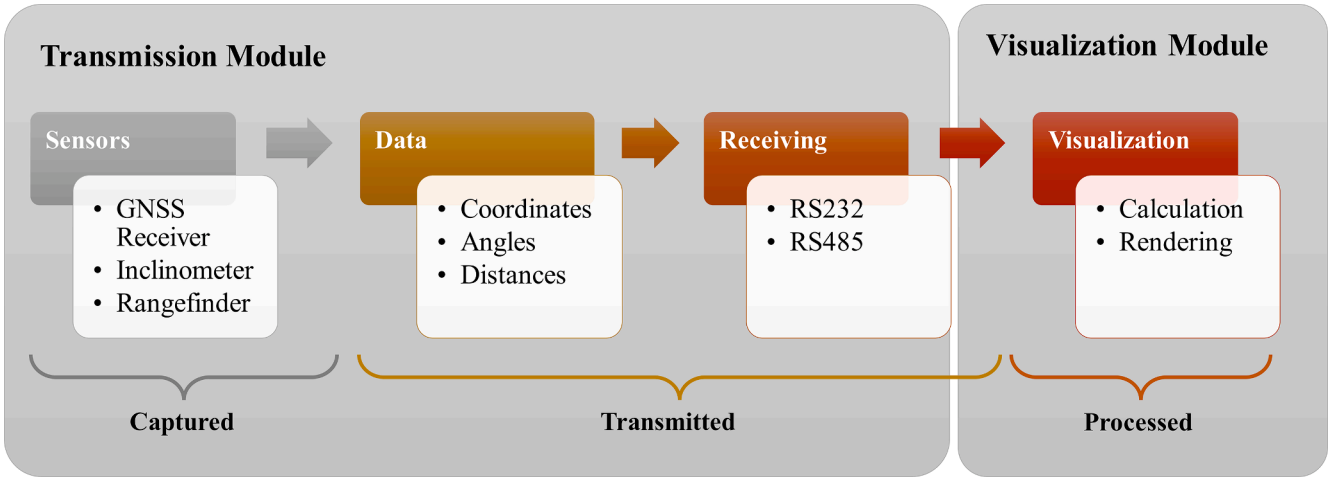


Fig. 5. Transmission module and visualization module in the data management module for capturing, transmitting, and data processing.

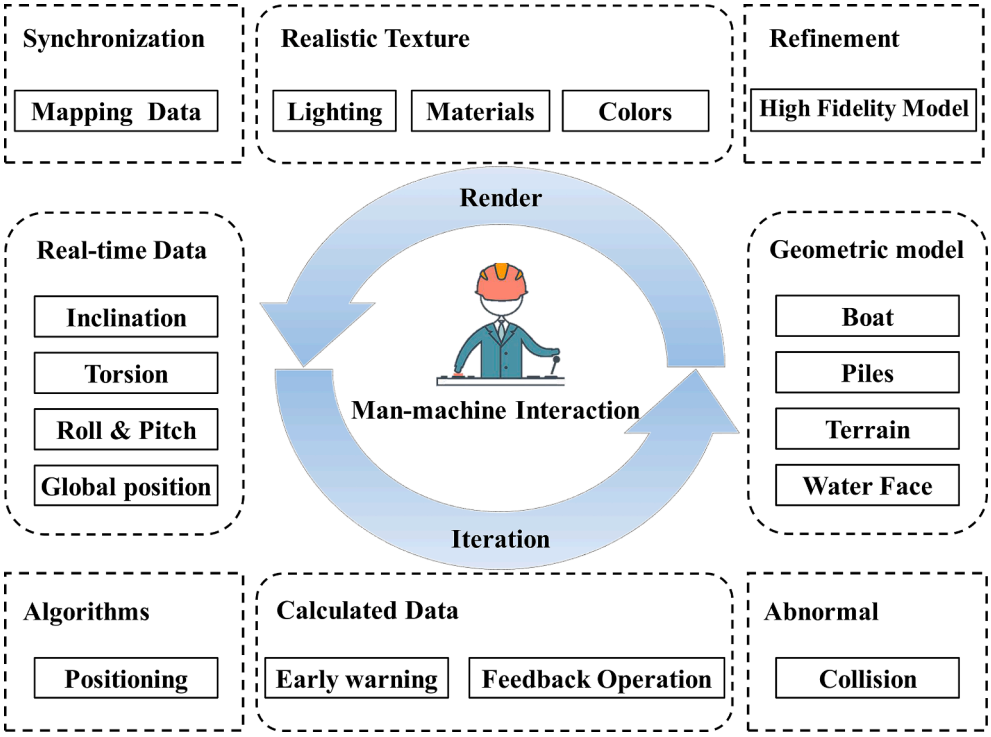


Fig. 6. Construction sense of collision early warning model for marine piling.

interaction encompasses the design of interfaces for human–computer interaction as well as the response to external input events in order to facilitate the navigation of a three-dimensional virtual environment. Additionally, the collision warning information data is updated in order to accurately reflect the real-world conditions of the construction site within the virtual environment.

The achievement of collision warnings is contingent upon the real-time mapping of the physical environment onto the virtual environment. Throughout the process of construction, the positions of both the pile and boat are established in accordance with the blueprint. This process involves notable dynamic changes in the attitudes of the pile as well as the spatial positions of both the pile and the boat. A dual-layer mapping approach has been established in order to accurately depict the dynamic behavior. Fig. 7 provides a depiction of the pile-boat mapping and the early-warning feedback mapping.

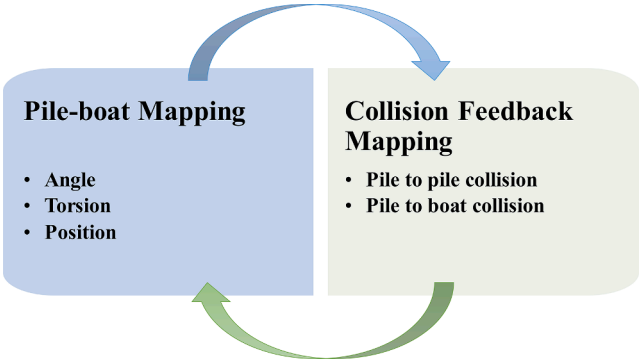


Fig. 7. Dual-layer mapping approach in the motion drive module.

3.3. Feedback module

The contents of the feedback service, which are essential for achieving the concept of "controlling the virtual with the real", are encompassed within the feedback module. The algorithm for pile positioning is utilized to compute the boat and pile attitude. This information is then employed to provide advanced warning regarding potential collisions between the under-construction pile and the piles that have been constructed. The feedback is relayed to the construction personnel, who can then adjust the pile attitude or move the boat accordingly. This process enables the control of virtual elements in reality. The procedure incorporates crucial components such as the pile positioning algorithm and the collision early warning algorithm, particularly in situations where pile positions are intricate.

3.3.1. Pile positioning algorithm

GNSS receivers are a crucial tool for acquiring precise location data during the process of pile driving in marine environments. It is designed to receive latitude and longitude coordinates, which can subsequently be transformed into specific engineering coordinates using seven parameters [14]. These parameters provide guidance to the boat, enabling it to navigate with precision within the construction area. Global positioning and navigation technology is now very mature. GPS, GLONASS, BeiDou, and Galileo are the four major GNSS in the world [34]. The GNSS receiver we use can receive all GNSS satellite signals and uses mature RTK technology. Therefore, in terms of positioning theory and hardware, its positioning accuracy can reach the millimeter level. In Sections 4.2 and Section 4.4, we list in detail the results of positioning calculations and collision calculations involved in actual piling projects. Ultimately, it is shown that global position data provide the required accuracies for the purpose of application. The use of a rangefinder or inclinometer can establish the appropriate orientation for the pile. The collection of measurement data in the physical environment is conducted in real-time, and the formation of twin data is achieved through the data management module. The motion drive module is responsible for achieving the rendering of piles.

As a result of the unique nature of pile construction, a hydraulic or diesel hammer is utilized to strike the pile body. After construction is completed, most of the piles are embedded in the soil. Consequently, the installation of a positioning device at any stage of the pile construction process is challenging due to cost considerations. The soil will squeeze or damage the device attached to the pile.

The task of determining the engineering coordinates of the point of intersection between the pile axis and the engineering design elevation plane, commonly known as the target point coordinates, is referred to as the positioning problem. Typically, as shown in Fig. 8 [3,11], a boat is equipped with two or three GNSS receivers. Two rangefinders are mounted either on the pile frame or the deck. The pile frame is equipped with an inclinometer. An O-XYZ Cartesian coordinate system is established, which is considered a ship-fixed coordinate system as it remains relatively stationary with the boat. The longitudinal axis of the boat is aligned with the OX axis, while the OY axis is perpendicular to the OX axis. Additionally, the OZ axis is perpendicular to both the OX and OY axes and intersects point O.

The pile positioning algorithm can be derived when the ship-fixed coordinates of the GNSS receiver, the center $(X_F, 0_F, Z_F)$ of the pile frame rotation axis, and the target point in the straight pile state $(X_S, 0, 0)$ are established. By utilizing two range finders, it is possible to accurately determine the inclination angle of the pile based on its distance. Alternatively, the inclination angle can be directly obtained through the use of an inclinometer. Subsequently, the stationary coordinates of the target point on the boat can be computed. By applying the conversion parameters that relate the ship-fixed coordinate system to the engineering coordinate system, it is possible to transform the ship-fixed coordinates of the target point into engineering coordinates. The specific calculation steps are shown in Fig. 9.

Fig. 10 illustrates the marine pile positioning process under the assumption that the pile frame remains stationary. Following eq. (1) is the calculation that is used to determine the coordinates of the target

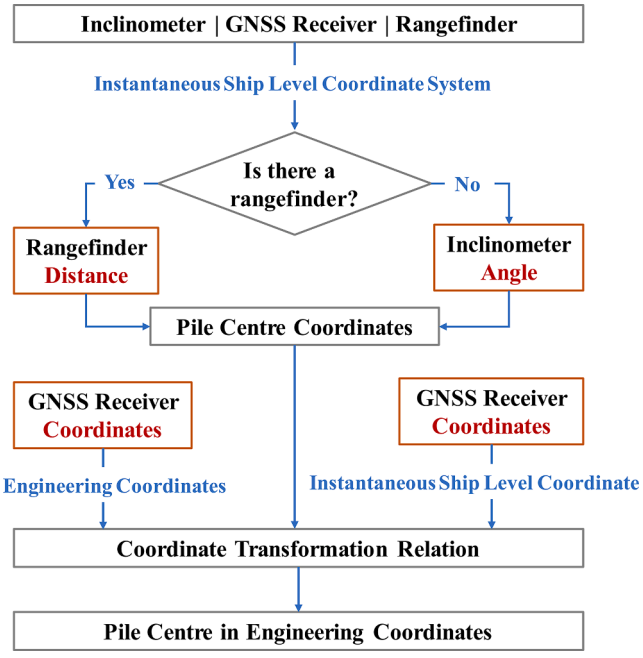


Fig. 9. Transformation steps of the ship-fixed coordinates of the target point into engineering coordinates.

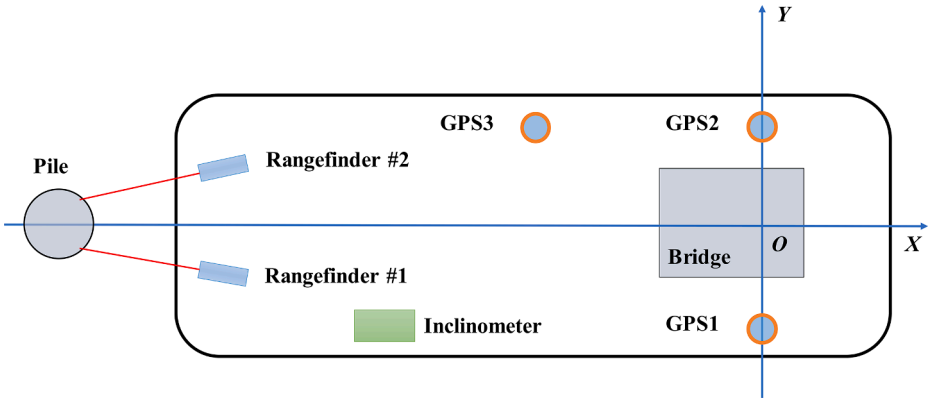


Fig. 8. Instrument installation arrangement and a ship-fixed coordinate system in a boat of marine piling.

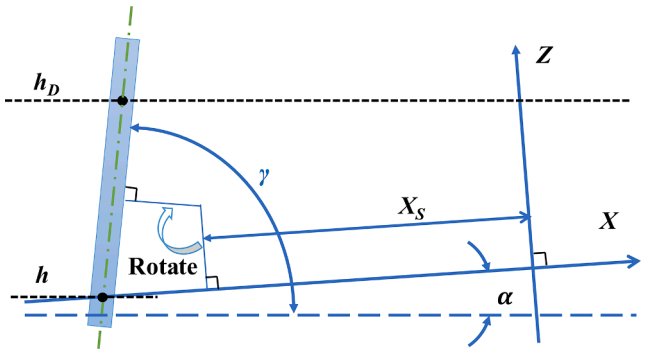


Fig. 10. Calculation diagram of the ship-fixed coordinates of the target point.

point (X_D, Y_D, Z_D) in the ship-fixed coordinate system. Therefore, α is measured by the inclinometer, h is calculated from the coordinates measured by the GNSS Receiver. Other parameters are quantitative and not measured by the instrument in real time.

$$X_D = X_F + \frac{X_S - X_F}{\sin(\gamma-\alpha)} - \frac{H_F}{\tan(\gamma-\alpha)} + \frac{h_D - h}{\sin\gamma}\cos(\gamma-\alpha), Y_D = 0, H_D = \frac{h_D - h}{\sin\gamma}\sin(\gamma-\alpha) \quad (1)$$

In the above equation, γ is the inclination angle of the pile, which can transform to the slope ratio. When it is a straight pile, $\gamma = \pi/2$. α is the trim angle of the hull. It is measured by an inclinometer. h_D is the elevation value of the design elevation plane. h is the elevation of the intersection plane of the pile axis and the OX axis in the engineering coordinate system.

3.3.2. Collision early warning algorithm

The DT-enable early collision warning system provides alerts for potential collisions between piles that are under construction and those that have already been constructed. Several collision detection methods exist. The advancement of computer graphics and building information modeling (BIM) technology in recent years has led to significant progress in the field of collision detection [32]. The utilization of BIM has been extensively employed in various situations, including metro and underground pipelines [33]. In ArcGIS and AutoCAD software, collision detection necessitates a pre-existing three-dimensional model. Currently, the correct positioning of piles claims a greater emphasis on collision avoidance, whereby there is no spatial overlap between the two. In the event that the distance between the two entities falls below the established safety threshold, a prompt notification is generated, furnishing details regarding the closest proximity point and offering corrective measures for distance regulation.

The algorithms for detecting collisions in spatial geometry exhibit a greater degree of complexity and diversity. The process of collision detection involves assessing the positional relationship between the center lines of two cylindrical objects, specifically determining whether they are co-linear, parallel, intersecting, or dissimilar. Additionally, the position of the standard plumb line and the center lines of the two pipes are taken into consideration. In cases where the central axis of a pipe deviates and the two pendants of its standard plumb line do not fall within the line segment concurrently, the algorithm may yield inaccurate results, indicating imprecision in the judgment condition. The precise characterization of the bus bar and the criteria for the junction of its ends are inadequate. Furthermore, there is a tendency to overlook computations or scenarios that have not been taken into account.

Most of these methods involve calculating cylinder-shaped spatial distances. Assuming diverse pile categories, such as square piles, are being contemplated within the simulated setting. Under such circumstances, the level of complexity is likely to escalate swiftly, rendering the task of devising a consistent and precise computational methodology arduous. Marine pile positioning has a vast spatial extent, and the number of piles designed is typically in the hundreds. An increase in the number of piles results in a significant rise in computational demands, thereby exerting a considerable processing burden on the data management and motion drive modules of the system. The operational efficiency of the DT system may be diminished as a result of this. Ensuring worldwide expeditiousness and optimal levels of exactitude and precision in collision detection is imperative. Consequently, the system incorporates a collision early warning algorithm that exhibits high efficiency. The procedural steps are as follows:

(1) Acquiring real-time twin data in the data management module. Pile's parameters consist of the engineering coordinates of piles that are currently under construction and those that have already been constructed, the inclination, the torsion angle, the type, the length, the width, and the radius.

(2) Utilizing parameters for virtual environment prejudgment. For round piles, the spatial geometry algorithm calculates the prejudgment distance as the distance between the axes of piles and the difference between the sum of the radii of two piles. Determine next whether the pile pair's axes are parallel. If the axes are parallel, the calculation procedure terminates, and the estimated distance is the actual distance. If it is not parallel and the pre-estimated distance is less than the predetermined safety distance, the pile pair may collide, and precise calculations must be performed. To calculate the pile pair axes, the vector method is used. Calculating the common normal vector of the two axes is the initial step. The projection length of the connecting line segment between any two points on the two axes is then calculated based on the common normal vector, i.e., the distance between the pile and the axis. Using the cross-product, the common normal vector can be found. Suppose that the vector connected by any two points is $\vec{m}$, and the included angle with the common normal vector $\vec{n}$ is θ, then the distance between pile and axis d_z is:

$$d_z = |\vec{m}||\cos\theta| = |\vec{m}|\frac{|\vec{m}\cdot\vec{n}|}{|\vec{m}||\vec{n}|} = \frac{|\vec{m}\cdot\vec{n}|}{|\vec{n}|} \quad (2)$$

Thus, the prejudgment distance is:

$$d_{pre} = d_z - (r_a + r_b) \quad (3)$$

where, r_a and r_b are the radii of each pile in the pile pair, respectively.

If it is determined to be a square pile or a hull, the convex hull of the square pile is used to represent the three-dimensional space equation of the pile:

$$A = \left\{ a \middle| a = \sum_{i=1}^{4} \lambda_i a_i, \sum_{i=1}^{4} \lambda_i = 1, \lambda_i \geqslant 0 \right\} \quad (4)$$

A is the dots set on the surface of a square pile or hull; a is a point of pile body; $a_i (i = 1, 2, 3, 4)$ represents the vertex coordinates of A; $\lambda_i (i = 1, 2, 3, 4)$ is a geometric parameter variable such as a shape parameter.

(3) Creating the distance function between pile pairs. The mathematical expression for determining the distance between two points is commonly referred to as the distance formula. The coordinates of the two points are represented by equation (4). The initial step involves computing the partial derivative of each variable in the distance function. Subsequently, a non-linear programming model is formulated, with the objective function being the minimization of the distance function and the constraint being the range of values for the independent variable in the pile space equation. The mathematical formula in question is presented below:

Assuming the calculation of the distance between two round piles. The prescribed values for u_a, u_b, v_a, v_b are determined based on the respective pile shape parameters. Subsequently, the constraint in question pertains to the box of the variable range. The objective function is derived by utilizing equations (2) and (3), resulting in:

$$\begin{aligned} &\min D = (x_a - x_b)^2 + (y_a - y_b)^2 + (z_a - z_b)^2 \\ &\quad \mathrm{gra} = \frac{d(D)}{du_a}, \frac{d(D)}{dv_a}, \frac{d(D)}{du_b}, \frac{d(D)}{dv_b}. \\ &[u_a]_{\min} \leqslant u_a \leqslant [u_a]_{\max}, [v_a]_{\min} \leqslant v_a \leqslant [v_a]_{\max} \\ &[u_b]_{\min} \leqslant u_b \leqslant [u_b]_{\max}, [v_b]_{\min} \leqslant v_b \leqslant [v_b]_{\max} \end{aligned} \tag{5}$$

where D is the objective function value; gra is the gradient function value; x_a, x_b, x_c, y_a, y_b, y_c are the coordinate values of the points on each pile in the pile pair, respectively. u_a and u_b are the included angles projected on the underside of a point on each pile face in the middle of a round pile. v_a and v_b are the length of a point on each pile face along the long direction of the pile.

If the distance between the round pile and square pile or round pile and hull is calculated, the constraint condition is a combination of a box constraint and a nonlinear constraint, and the objective function is as follows:

$$\begin{aligned} &\min D = (x_a - x_b)^2 + (y_a - y_b)^2 + (z_a - z_b)^2 \\ &\quad \mathrm{gra} = \frac{d(D)}{d\lambda_i}(i = 1,2,3,4), \frac{d(D)}{du}, \frac{d(D)}{dv} \\ &\quad s.t. \sum_{i=1}^{4} \lambda_i = 1, \lambda_i \geqslant 0, \ i = 1,2,3,4 \\ &\quad [u]_{\min} \leqslant u \leqslant [u]_{\max}, [v]_{\min} \leqslant v \leqslant [v]_{\max} \end{aligned} \tag{6}$$

where u and v are geometric parameter variables such as the shape parameter of the pile-to-central round pile.

If the distance between square piles is calculated, the constraint condition is a nonlinear constraint, and the objective function is:

$$\begin{aligned} &\min D = (x_a - x_b)^2 + (y_a - y_b)^2 + (z_a - z_b)^2 \\ &\quad \mathrm{gra} = \frac{d(D)}{d\lambda_i}, \frac{d(D)}{d\sigma_j}(i,j = 1,2,3,4) \\ &\quad s.t. \sum_{i=1}^{4} \lambda_i = 1, \sum_{j=1}^{4} \sigma_j = 1 \\ &\quad \lambda_i \geqslant 0, \ \sigma_j \geqslant 0 i, (j = 1,2,3,4) \end{aligned} \tag{7}$$

where, $\lambda_i (i = 1,2,3,4)$ and $\sigma_i (i = 1,2,3,4)$ are geometric parameter variables such as the square pile or hull shape parameters, respectively.

(4) The non-linear programming problem is addressed through the utilization of either the sequential quadratic programming algorithm or the box-constrained quasi-Newton algorithm. This enables the determination of the minimum distance between pairs of piles that are at risk of colliding. If the minimum distance is less than the predetermined safety distance, a collision warning will be issued in the data visualization module of the data management module. Simultaneously, the feedback pertaining to construction is furnished within the module for instructional feedback.

3.4. System management module

Data transmission, interface calling, signal transmission, etc. are all processed between modules via the system management module. It mainly has three functions, namely real-time data flow update, file and log management, and inter-module coordination. Fig. 11 shows these three functions and the specific contents they contain.

For real-time data flow updates, the data management module sends requests to the data transmission module, and the system management module provides a response based on the data management module's mapped data list to obtain data in real time. At last, the data is updated in the request-response cycle, allowing the data visualization module to update the appropriate graphical user interface.

For file and log management, first, implement the hardware device data archiving of the data management module. These data are original monitoring data during construction. They are first-hand information on the construction site directly measured by various monitoring instruments. Then, the virtual environment data archiving of the motion drive module is implemented. The evolution process of the virtual environment is recorded to facilitate subsequent analysis and tracing. Finally, the two algorithm calculation processes and result archiving in the feedback module are implemented. This is the purpose and core of the entire piling construction, and it is also the most valuable data in the DT system proposed in this study.

For inter-module coordination, in order to achieve the normal operation of various functions, coordination between modules is necessary. In particular, the DT-enabled collision eraly warning system uses a method that encompasses four modules and sets up relatively independent modules based on the topological relationship of modular subsystems, allowing for rapid updates to the iterative module subsystem based on the needs of field operations and access to the real-time data trigger procedure. The functionality of other modules will not be adversely affected by maintenance on any one module's subsystem. For instance, it is possible to iterate the data visualization module's function while maintaining reliable motion driving and data transfer. Therefore, the system management module must undertake the coordination work between modules.

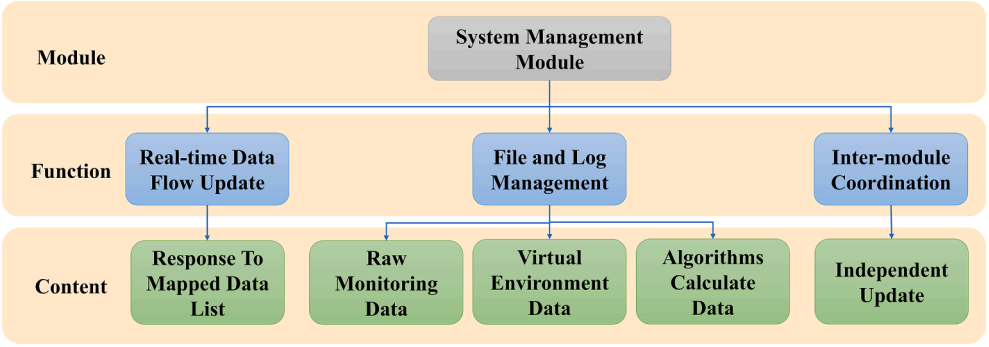


Fig. 11. Functions and specific contents of the system management module.

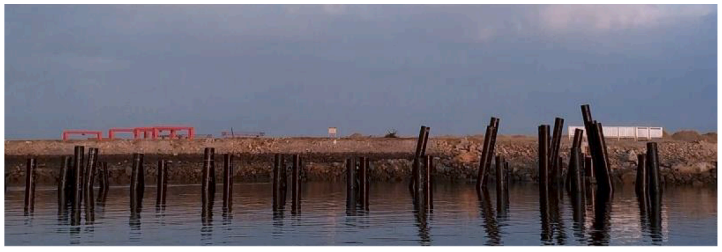
4. Case study

4.1. Project overview and procedure introduction

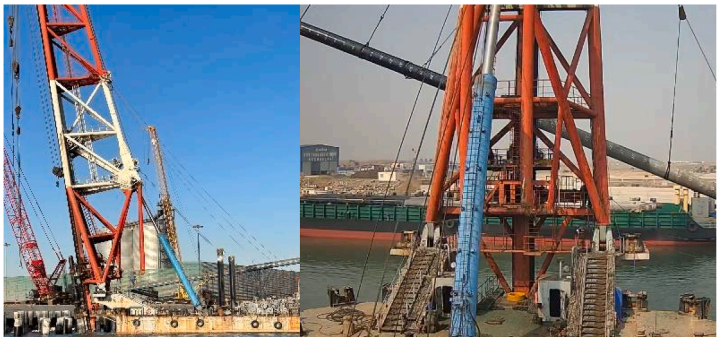
Two 10,000-ton general berths are planned as part of a wharf project to accommodate the rising need for shipping components for wind power generators, as well as steel, minerals, and other general cargo. The overall dimensions are 337 m long, 55 m wide, and 36 m wide at the bearing platform. Round piles with a diameter of 1,200 mm are used for the pile foundation. There are 10 transverse rows of piles, each having five straight piles and five inclined piles ranging in length from 54 m to 60 m, with a transverse spacing of 6 m between each row. A typical row frame has five straight piles, two 15:1 inclined piles at an angle of 3.8°, two 5:1 inclined piles at an angle of 11.3°, and one 4.5:1 inclined pile at an angle of 12.5°, with a bottom elevation of around −50.61 m to −54.85 m. The pile weighs around 75 metric tons on average. A complicated and dense cap is formed by several of the inclined piles. Fig. 12 (a) shows the three-dimensional drawing. Fig. 12 (c) shows a part of sectional view of the wharf's structure. The 2.5 m thick superstructure is made of cast-in-place concrete piers.

The Rushan Shipbuilding Co., Ltd. built Yunzhuang No. 1 boat is being used for the project. The maximum load-bearing capacity while the pile is straight is 80 t, and the main body measures 64.6 m in length and 26.00 m in width. The amplitude range of the pile frame is ± 25°. The full pile length is 70.00 m, and the maximum pile diameter is 2500 mm for the building. Pile frame load-bearing capability and construction requirements dictate a maximum variation range of ± 11° for pile frame inclination angle during the construction of 75-ton inclined piles, which cannot reach the design value of 12.5°. Consequently, construction errors are inevitable.

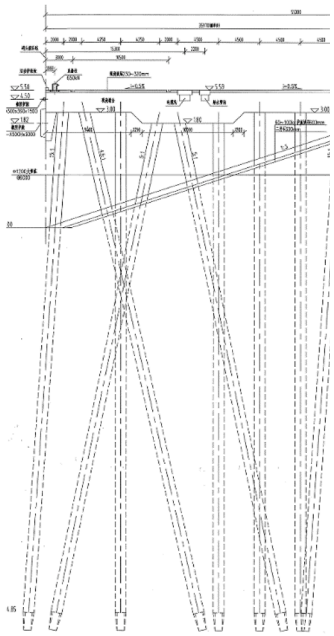
The DT-enabled collision early warning system is developed using Visual Studio 2022 communite version and Helix Toolkit as the three-dimensional geometric model image display engine, as per the previous five-dimensional model. Visual Studio 2022 is an integrated development editor developed by Microsoft. It is a rich development tool. Helix Toolkit is a collection of three-dimensional components for .NET Framework [35]. It provides many high-level API interfaces, making development easy. Consequently, the virtual environment of marine piling is established utilizing the .NET 6.0, which is a free, open-source, cross-platform framework. The combination of this framework, Visual Studio 2022, and the Helix Toolkit allows us to build the DT-enabled system quickly and cost-effectively. Fig. 13 shows the comprehensive functions of the system. The static information of the virtual environment is extracted from the boat shape and construction pile information files before actual construction commences. The three-dimensional geometric model of the pile and the boat shape is automatically generated by the virtual environment upon input of the control point and the pile information file, as illustrated in Fig. 13 (b). Fig. 13(c) shows a 3D virtual scene that can be manipulated via human–computer interaction to enable the user to view the real-time status of construction from various perspectives of the three-dimensional view and the plan layout. The information pertaining to the current position of the construction pile, torsion angle, and inclination is displayed on the left-hand side. The preliminary alert data provides insight into the likelihood of a collision between the present pile and the constructed piles following the completion of construction based on the current boat position and pile inclination angle. This information is updated in real-time and synchronized with the virtual environment displayed on the right-hand side. Real-time data is utilized to display instrument monitoring information in textual format in the lower section of the system. Additionally, twin models of the pile, boat, water surface, and terrain are presented in the three-dimensional view. The plan layout displays pertinent information regarding the placement of piles in relation to the engineering design elevation plane. The feedback operation beneath the system is based on the distance between the construction pile position and the design pile position, as well as early warning information. The system provides information regarding the current orientation of the boat in terms of fore and aft, left and right, and boat direction, with reference to the bow view. Additionally, it utilizes virtual environment simulation to guide the positioning of piles and achieve real-time visualization.



(a) Actual construction in the physical environment.

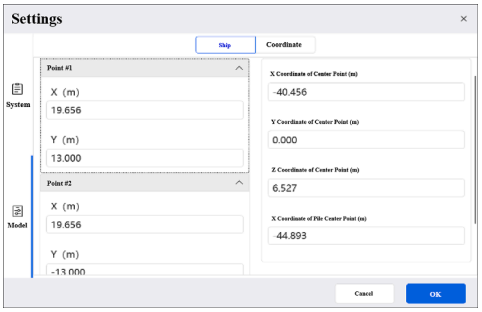


(b) Yunzhuang No. 1 boat.

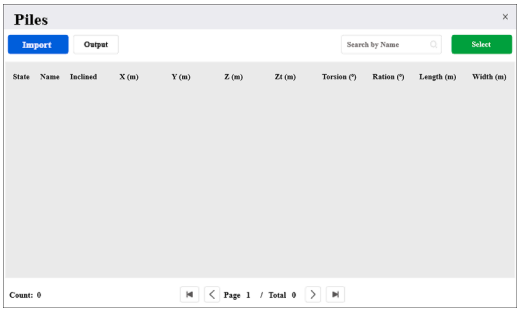


(c) Structural section wharf.

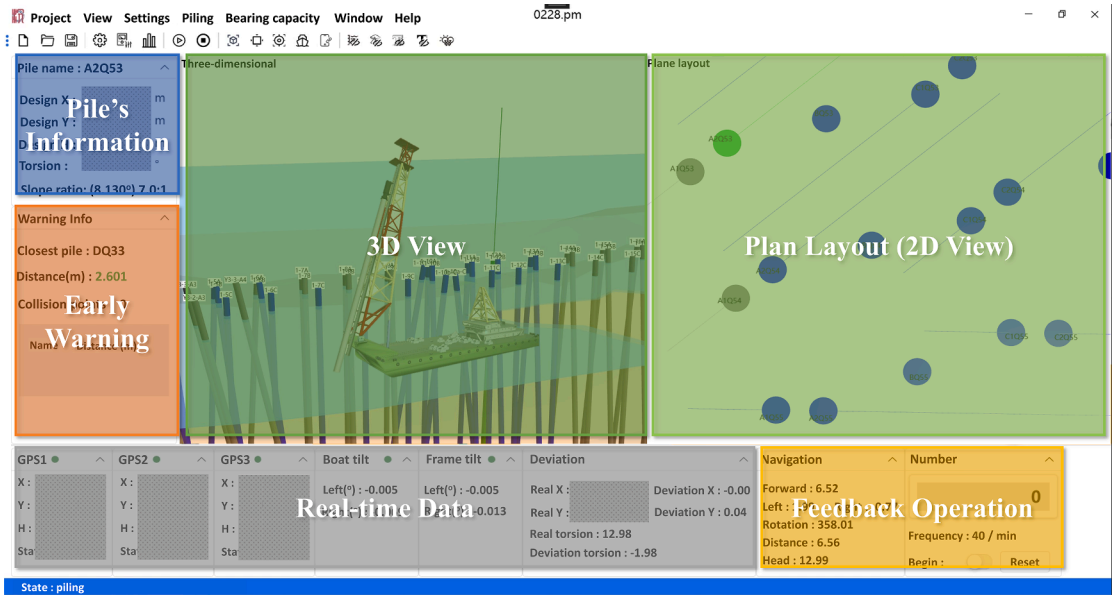
Fig. 12. Project overview. (a) practical construction site in the physical environment, (b) Yunzhuang No. 1 boat, and (c) wharf structure section of a complex pile position consisting of some inclined piles.



(a) System parameter setting interface.



(b) Pile information file inputting interface.



(c) Essential components of the DT-enabled collision early warning system.

Fig. 13. Comprehensive functions of DT-enabled collision early warning system. (a) Interface of boat shape parameters, (b) pile information file inputting interface, and (c) essential componets of the system.

4.2. *Positioning accuracy*

We deployed this system in practical on-site construction and obtained process data. Differences in post-construction coordinates for 15 inclined and 15 straight piles are listed in Tables 1 and 2. The post-construction coordinates are measured by the construction crew using a total station. These coordinates represent the actual geographical location of the completed pile. The difference between it and the design coordinates represents a construction error. The difference between the coordinates of A2Q58 in the inclined pile is greater than 20 cm, which exceeds the maximum deviation of ± 15 cm allowed by the specification. Additionally, the X-engineering coordinates of BQ48 and the Y-engineering coordinates of BQ41 in the straight pile exceed the maximum deviation of ± 10 cm allowed by the specification, indicating that there is a significant deviation in the actual construction. By comparing the system positioning coordinates to the actual geographical location, it can be seen that the errors are within 0.1 cm, which means that the pile positioning algorithm of the system feedback module is doing very well.

Table 1
The difference between the design and actucal coordinates and the difference between the proposed system and actucal coordinates of inclined piles. The actual coordinates are measured by the construction crew with a total station. (unit: m).

Pile	Design to actual		System to actual	
	X	Y	X	Y
A2Q58	**0.280**	**0.138**	−0.003	0.001
A2Q59	0.036	−0.007	−0.004	−0.005
C1Q52	0.158	0.063	0.007	−0.001
C1Q57	0.097	0.029	0.002	−0.006
C1Q58	0.049	−0.004	−0.002	−0.002
C2Q48	0.117	−0.019	0.007	−0.009
C2Q49	0.044	0.099	0.005	−0.010
C2Q50	0.179	0.095	0.010	0.005
C2Q51	0.167	−0.114	0.010	−0.008
C2Q52	0.113	0.060	0.000	−0.007
C2Q53	−0.011	−0.051	0.010	0.006
C2Q54	0.169	−0.002	0.002	0.008
C2Q55	0.138	−0.057	−0.006	0.010
C2Q56	−0.077	0.061	−0.005	−0.008
C2Q57	−0.014	0.025	−0.003	0.001

4.3. *Collision early warning under complex pile position*

Complex pile position is a common occurrence in wharf construction projects. Here, we use A2Q53 to illustrate the collision early warning

Table 2
The difference between the design and actual coordinates and difference between the poposed system and actual of straight piles. The actual coordinates are measured by the construction crew with a total station. (unit: m).

Pile	Design to actual		System to actual	
	X	Y	X	Y
BQ39	0.057	0.084	−0.003	0.004
BQ40	−0.125	−0.108	0.000	0.002
BQ41	0.079	**0.143**	0.005	−0.009
BQ42	0.070	0.003	−0.005	0.005
BQ44	0.144	0.085	−0.001	0.001
BQ46	0.096	−0.079	−0.002	0.007
BQ47	0.197	−0.046	−0.001	0.009
BQ48	**−0.203**	0.043	0.003	−0.003
BQ49	−0.035	−0.119	0.008	−0.003
BQ50	−0.048	0.049	−0.008	−0.007
BQ51	−0.137	−0.002	0.005	−0.007
BQ52	0.033	0.127	−0.004	0.003
BQ53	−0.011	0.016	−0.003	0.007
BQ54	−0.025	0.055	0.006	−0.006
BQ55	−0.099	−0.030	−0.006	0.004

shown in Fig. 14 (a). The A2Q53 is an inclined pile with a slope ratio of 4.5:1 and a torsion angle of 20°. From the design drawing, it appears that the arrangement of surrounding piles is suitable for field construction, as there is sufficient space for pile driving. However, the spatial position must be considered during field construction. C1Q52 and BQ53 may collide with them. In the design phase, the closest distance between A2Q53 and other piles is approximately 0.74 m. Due to a construction error in pile C1Q52, which is documented in Table 1, this distance was reduced to 0.65 m, resulting in a small distance between piles. The risk of collision can be clearly indicated in the system.

When the collision distance is less than the predetermined safe distance, the system will send a warning consisting of text and graphics about the virtual environment. The system has determined that a safe distance of 3.0 m should be maintained and that the pile name is A2Q53. All of the nearby piles are finished. Fig. 14 (d)(e) shows the early warning information and a three-dimensional view of the system interface. When collisions occur, the system will pop up a warning window to remind workers to operate with caution. After confirming the information, it will not pop up again to avoid interference with the operation. Warning details will be displayed on the system's early warning panel. The system provides the following warning information: the nearest pile's name, BQ53; the nearest distance, 0.567 m; the number of collision points, 5, and a table of all piles that may collide at a given safe distance. The unconstructed mass is depicted in gray in the three-dimensional view. The constructed piles are colored blue, the intended pile positions are colored green, the current pile is colored red, and the hull is colored orange. The possible collision pile pairs are connected by yellow line segments to signal the collision path.

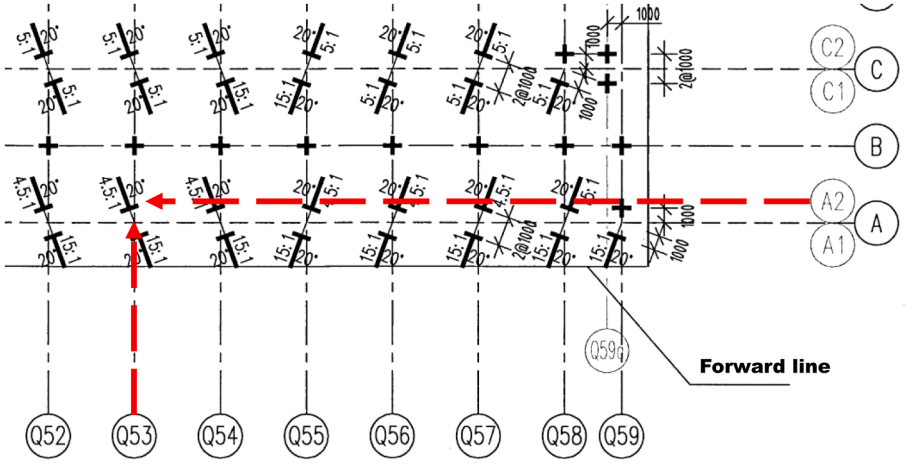
4.4. Calculation performance of collision early warning

After the system was put into operation, it showed good functionality and real-time performance. Through actual testing and pressure testing, and under the condition of real-time warning, the latency between the virtual and physical environments is approximately 0.2 s, which satisfies the requirements of collision warning and monitoring in the actual pile positioning construction. Thus, the viability and efficacy of a collision early warning system facilitated by a DT are substantiated. In order to more effectively showcase the rationality and efficacy of the collision early warning algorithm, the boat was gradually relocated within the virtual environment in accordance with the engineering conditions. The range of motion encompassed the entirety of the construction site in order to replicate conditions that would necessitate collision warnings. Table 3 presents a part of the pile data, encompassing the initial design data for unconstructed piles and the actual pile position data for constructed piles. The non-linear programming model was utilized to compute the design data of 606 pile groups based on the collision early warning algorithm. There were 367,235 calculations performed. The minimum distance between pile pairs is between pile No. 33 and pile No. 66, which is 30.85 cm. Windows 11 is the operating system for the hardware environment of the test. The central processing unit operates at a frequency of 2.90 GHz, while the memory capacity is 16 GB. A single collision calculation for 606 data sets takes an average of 0.11 ms. The speed that has been computed is sufficient for the purposes of real-time collision monitoring and early warning during pile positioning.

4.5. Further discussion

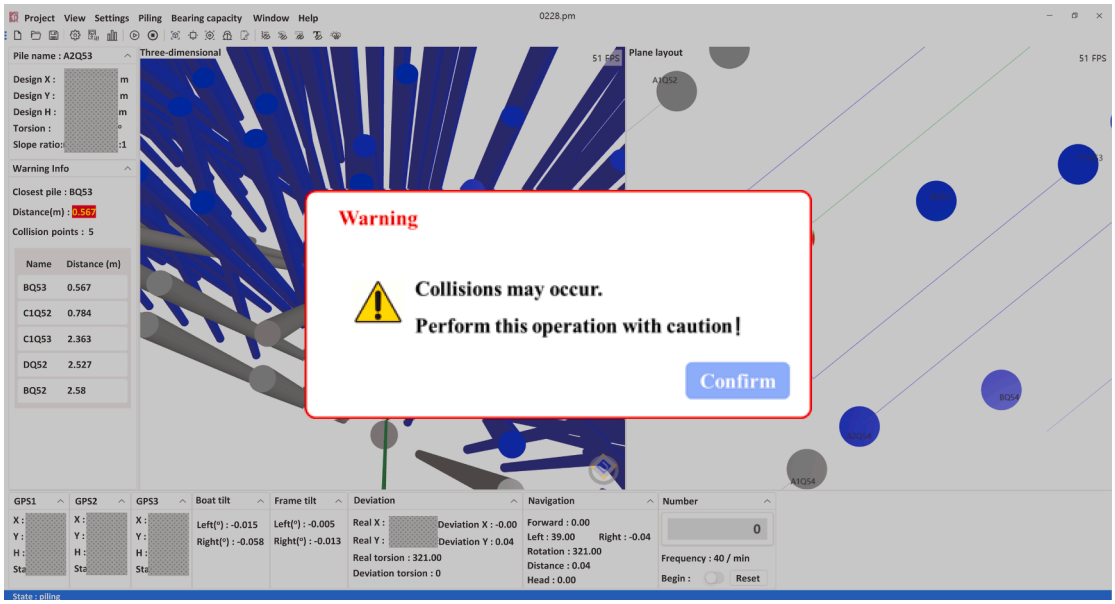
4.5.1. Control between the real and the virtual

Construction personal can combine actual construction and DT to adjust the position of boat and pile in the physical environment. This

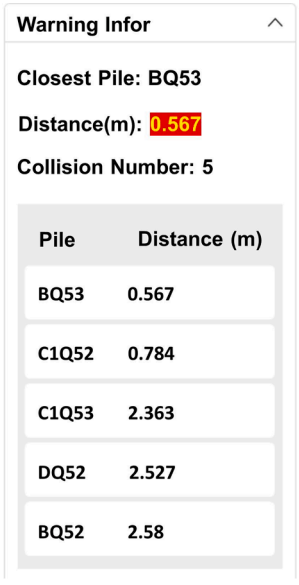


(a) 2D arrangement of pile A2Q53 in design drawing.

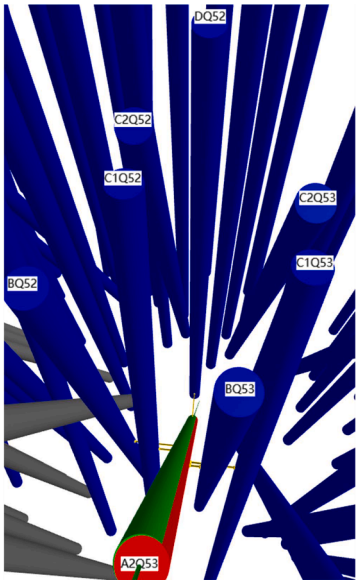
Fig. 14. Real-time collision warning and three-dimensional view warning information under the current inclination and torsion in the virtual environment. (a) two-dimensional arrangement of A2Q53 in design drawing, (b) warning window of collision possibilities. It only appears for the first time when collisions may occur, (c) warning information in brief and table of piles that may collide, (d) three-dimensional view warning corresponding to warning information, and (e) closely observe the collision warning information in three-dimensional view and mark it with numbers in sequence.



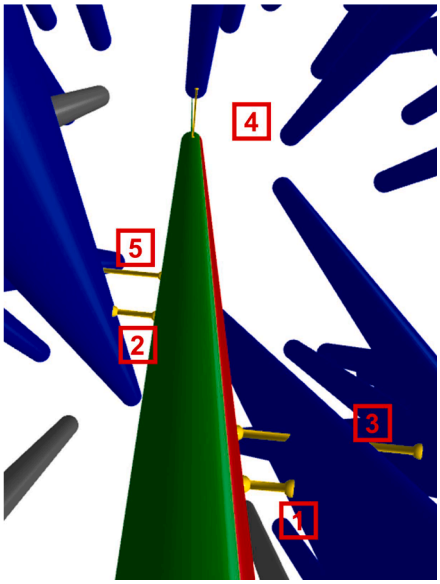
(b) Warning window of collision possibilities. It only appears for the first time when collisions may occur.



(c) Warning information in brief and table of piles that may collide.



(d) 3D view warning corresponding to warning information.



(e) Closely observe the collision warning information in 3Dview and mark it with numbers in sequence.

Fig. 14. (*continued*).

enables real-time mapping. That is the concept of controlling the virtual with the real. Simultaneously, based on the alterations observed in the twin data, the system expeditiously computes the location of the pile and collision warning data to provide risk alerts during operation and furnish feedback guidance to avert collisions. The system furnishes the scientific foundation for construction personnel to render effective decisions in managing the interface between the physical and virtual realms. The efficient and safe construction of marine piling is made possible through the full demonstration of controlling virtual with real and controlling real with virtual and real-time mapping.

Table 4 enumerates the precise functionalities of the system that have been augmented through the utilization of DT technology to facilitate piling. The essence of a DT is its ability to manipulate the actual world with the virtual world and vice versa. In addition to incorporating the existing capabilities of the prior piling position system, which encompasses the analysis and processing of sensor data to derive positioning information, the current system also encompasses novel functionalities that were previously unattainable. The visual

Table 3
Design data of piles in a wharf project.

Pile number	Shape	Length (m)	Width (m)	Torsion (°)	Slope Ratio	Coordinate X (m)	Coordinate Y (m)
1	round	39.5	0.8	244.6	6	4510447.28	412499.32
2	round	39.5	0.8	244.6	6	4510453.18	412498.24
3	round	39.5	0.8	244.6	6	4510459.08	412497.16
4	round	39.5	0.8	244.6	6	4510464.99	412496.08
5	round	39.5	0.8	244.6	6	4510471.38	412494.91
……	…	……	……	……	……	……	……
33	round	39.5	0.8	64.6	4.5	4510453.4	412499.5
……	…	……	……	……	……	……	……
66	round	39.5	0.8	334.6	6	4510454.18	412498.7
……	…	……	……	……	……	……	……
605	round	47	0.8	346.6	5	4510628.31	412488.64
606	round	47	0.8	169.6	5	4510624.37	412489.36

Table 4
Comparison between our DT-enabled collision early warning system and traditional positioning system.

Ability to assist with construction		Our system with DT	A traditional system without DT
Controlling the real with the virtual	Sensing data analysis processing	√	√
	Closed sensing data flow with early warning information	√	×
	Pile position in two-dimensional and three-dimensional space	√	×
	Visual display of collision risk	√	×
Controlling the virtual with the real	Display the degree of risk with distance as a quantitative standard	√	×
	Boat's movement feedback	√	×

representation of collision risk is specifically exhibited in the piling position.

4.5.2. *Purpose of digital twin in the piling construction*

DTs are widely used in the manufacturing. They have been established for the whole product lifecycle [36] and workshops [37]. It provides rich experience and cases for the application of DTs in the civil engineering. The piling construction belongs to the civil engineering. The standard for DT in the manufacturing are established. Kritzinger et al. [38] summarized reported researches in the manufacturing and divided those DTs into digital model, digital shadow, and DT. This classification is based on the degree of data integration. Specifically, if data flows automatically in both directions, it is a DT, but not vice versa. But the manufacturing is different from the civil engineering.

(1) The research objects are different. The research objects in the manufacturing are mostly products and product-related processes, while research objects in the civil engineering are mostly engineering structures and construction processes such as piles and boat in the piling construction.
(2) The goals are different. The manufacturing hopes to completely replace labor in the entire process, while the civil engineering cannot full automation. From our point of view, civil engineering is closely related to human life safety. Once problems occur, such as bridge collapse, uneven settlement of houses, or wharf structure failure, it will cause immeasurable economic losses and even loss of life. Therefore, it must involve manual participation. Otherwise, we will not be able to find someone to take responsibility if something goes wrong since there is no mature law that defines where this responsibility belongs. Ergo, the DT is to assist construction personal in decision-making.

The DT is designed to assist piling construction in this study. The DT and its architecture proposed in this study are in line with the views of Grieves [16] and Tao [20–22], notwithstanding the classification based on the degree of data integration are not suitable for the current piling construction.

5. Conclusions and future work

Marine piling construction uses two-dimensional representation that lacks three-dimensional spatial information, and the pile position and attitude under the water surface are invisible during construction. In particular, actual construction machinery limitations and the physical act of construction cause construction to deviate from design. These problems increase the risk of collision. The application of DTs in marine piling demonstrates its powerful capabilities. DTs solve the above-mentioned objective problems of two-dimensional representation and invisibility under the water through real-time three-dimensional mapping of physical space. The risk of collision is mitigated through real-time collision early warning. A DT-enabled five-dimensional model and four system modules that can be separately maintained and improved are presented to realize early collision warnings between under-construction piles and constructed piles. Piles are main DT objects. The relationship between the modules, their specific implementation techniques, and their algorithms are elaborated upon. This system was deployed in practical projects and was supported by on-site construction data. The following are their respective innovations and conclusions:

(1) The physical environment, virtual environment, feedback services, twin data, and connections comprise the five-dimensional model. GNSS receivers, inclinometers, rangefinders, terrain, water, piles, and boat are included in the physical environment. Physical entities are virtualized within the virtual environment. Feedback services include both operational and functional services. Physical environment data, virtual environment data, and computational feedback service data comprise the twin data. Connections are crucial for realizing the real-time mapping of the DT's various components for marine pile positioning. A real-time mapping of the virtual world to the actual environment with a latency of less than 0.2 s was produced in a practical wharf project.
(2) The DT system was deployed at the practical construction site. On-site data results show that for the inclined pile, the difference exceeds the ± 15 cm allowed by the specification, and for the straight pile, the deviation exceeds the ± 10 cm allowed by the specification. However, the DT system positioning coordinates show errors within 0.1 cm. It is closer to reality, which indicating

the pile positioning algorithm of the system feedback module is performing well.

(3) The algorithm for implementing the feedback service in the feedback module has a comprehensive explanation. The formulation of the pile positioning algorithm is presented, along with its derivation. A collision early warning algorithm is constructed by integrating algorithms for spatial geometry, collision detection, and non-linear programming intelligence. It is validated by the average collision detection rate of 0.11 ms for a set of pile pairs.

(4) The positioning accuracy of the proposed DT-enabled collision early warning system meets engineering requirements. The collision information can be accurately, intuitively, and quickly prompted. It shows that the overall method flow of the system can smoothly assist actual project construction. The system is developed using .NET 6.0 and compatible technologies. It is decomposed into four modules that can be maintained independently. Therefore, it can be easily transplanted and applied to similar projects, such as land pile driving.

It should be noted that the system relies only on a virtual environment display, which might cause workers to lose faith and focus on their own knowledge and expertise rather than on real situations. Future work might improve the information density by either increasing the realism of the virtual environment rendering or employing human–machine-compatible reality augmentation technologies to add virtual information directly to the real world. Also, we will focus on creating user-centric experiences that improve the availability and efficiency of the system. Therefore, the system will be updated to behavioral DTs to understand and optimize human interactions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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